

NBA Player Salaries and Market Value

Capstone Final Project Report

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Problem Statement

NBA front offices operate under a hard salary cap, which means every dollar committed to one player is a dollar that cannot be spent on another. Identifying which players are paid in line with their production, and which are overpaid or underpaid, is key to building an efficient roster.

The question driving this project is: based on 2025-26 regular season performance, how well can a player's salary be predicted from measurable production, and which players fall furthest from their predicted value? The target audience is an NBA front office and general managers who can use the results to guide contract decisions, trades, and roster construction.

Data

The dataset combines three sources for the 2025-26 NBA season. Player season totals and biographical data such as height, weight, and wingspan were pulled from the `nba_api` LeagueLeaders endpoint. Player salaries were collected from Basketball-Reference.com. Collective bargaining agreement constants, including the 2025-26 salary cap of 154.6 million dollars and the minimum and maximum salary tiers by years of service, were added to provide contract context.

After merging the three tables on player id and normalized player names, the raw dataset contained 582 players. The main predictors used for modeling were:

- Per 36 minute rate stats – points, offensive and defensive rebounds, assists, steals, blocks, turnovers, field goal attempts, three point attempts, free throw attempts, and personal fouls
- Shooting percentages – field goal, three point, and free throw percentage
- Minutes per game – average minutes played, a proxy for role and coach trust
- Games played – a measure of availability across the season
- Years of experience – seasons of service, which also sets contract eligibility under the CBA
- Age – player age in years

The target variable is `log_salary`, the natural log of 2025-26 salary. Physical measurements were reserved for the clustering step rather than the salary model.

Data Wrangling

Considerable effort went into producing one clean record per player. The salary table from Basketball-Reference contained repeated header rows, dollar amounts stored as text strings, and duplicate entries for traded players. These were cleaned by removing header rows, converting salary strings to numbers, and keeping the highest salary row for each player.

Six players were missing a weight value. These were almost all two way and G League contracts with limited minutes, so the median weight was imputed for them. A larger group of 113 players had no salary on record, again mostly two way contracts. These players were excluded from the salary model but retained for clustering, where salary is not required.

Exploratory Data Analysis

Exploratory analysis focused on the distribution of salary, the relationship between production and pay, and deciding which players to include in the model.

Salary is heavily right skewed. A large share of players sit near the lower end of the pay scale on rookie deals, veteran minimums, or mid level exceptions, while a small number of stars earn the maximum. Because of this skew, the natural log of salary was chosen as the modeling target so that the distribution is closer to symmetric and the largest contracts do not dominate the fit.



Salary is right skewed both before and after the minutes filter, which motivated a log transformed target.

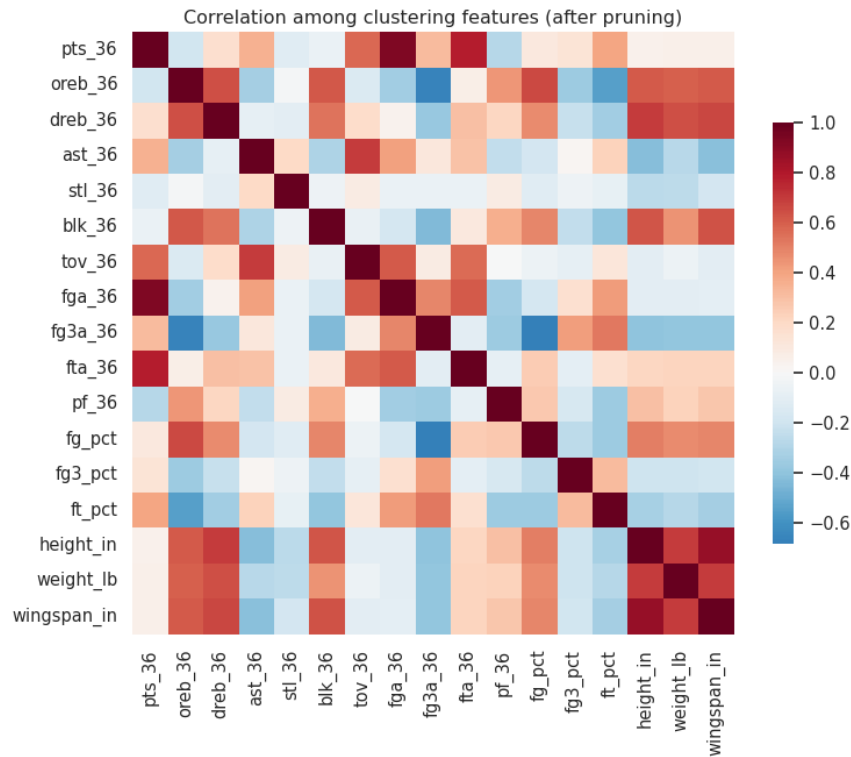
Scatterplots of salary against each statistic showed that offensive volume measures such as field goals made, field goal attempts, free throw attempts, and points had the strongest linear relationships with pay, while defensive counting stats like rebounds, blocks, and steals were weaker. In short, the market rewards offense more visibly than defense. Shooting percentages showed no clear linear relationship with salary on their own, since low salary players can post good percentages on low volume.

A minutes cutoff was applied to reduce noise from players with very limited playing time. Players with fewer than 500 minutes were removed, leaving 379 players for analysis. The high scorers below the cutoff were injured stars, which confirmed that the cutoff was removing small sample noise rather than genuine contributors.

Preprocessing

The counting statistics were converted to a per 36 minute basis so that rate of production is compared rather than raw totals, which would otherwise reward players simply for playing more. Total rebounds were left out of this step because they are the exact sum of offensive and defensive rebounds and would duplicate information.

All predictors in the salary model are numeric, so no categorical or dummy encoding was required. Features were standardized to z scores using StandardScaler, which is necessary for the distance based clustering and helpful for the linear model. A correlation review confirmed expected overlaps, such as field goal attempts with points and the physical measurements with one another, which informed feature selection.

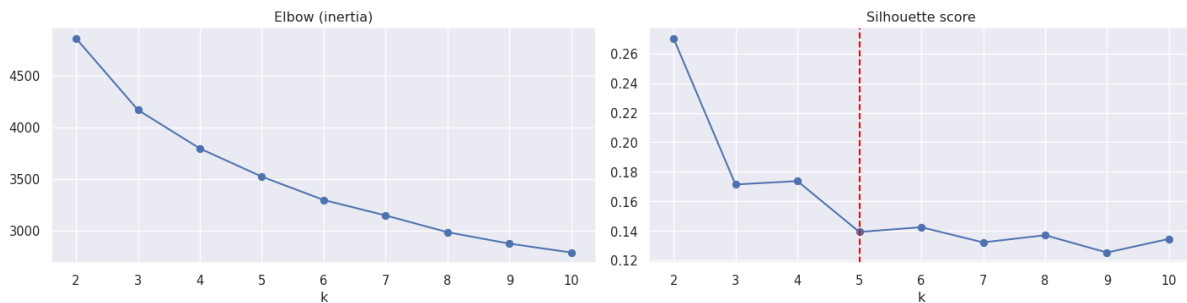


Correlation among clustering features. Field goal attempts and points overlap, as do the physical measurements.

Finally, the 16 players within the 379 that still lacked a salary were dropped from the modeling set, leaving 363 players. This set was split 80 percent for training and 20 percent for testing, giving 290 training players and 73 test players.

Player Archetypes

Before modeling salary, k means clustering was used to group players into archetypes based on their per 36 production, shooting, and physical measurements. The number of clusters was set to five based on the elbow and silhouette curves, where the improvement flattened after k equals four to five.

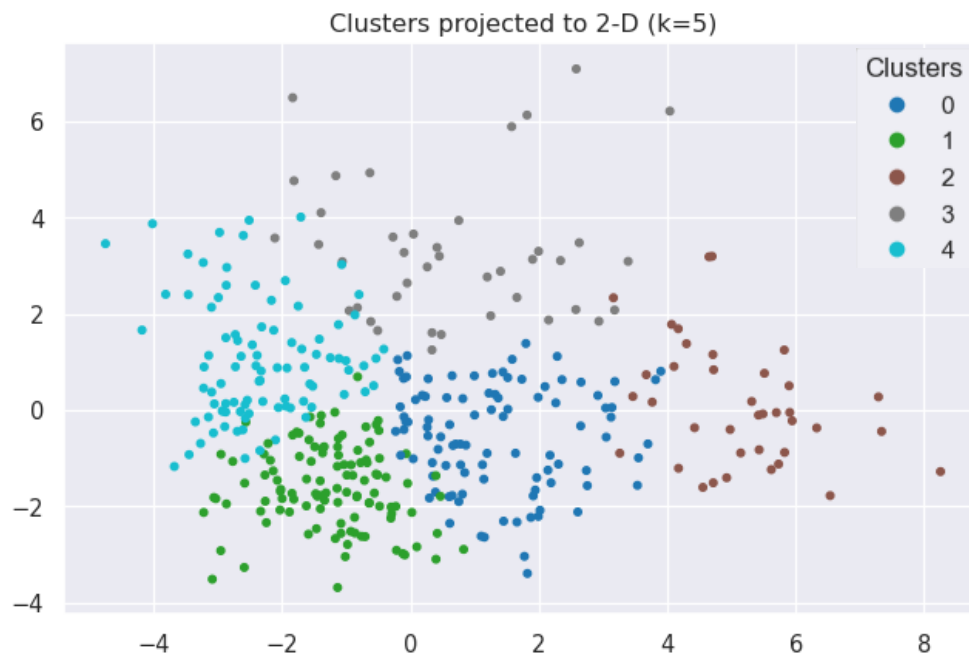


Elbow and silhouette scores used to select five clusters.

The five archetypes, with their average salary, were:

- Cluster 3 - Elite Superstar Frontcourt (41 players, about 36.1 million dollars) – the highest paid group of maximum contract forwards and bigs

- Cluster 4 - Elite Scoring Guards and Playmakers (91 players, about 17.7 million dollars) – high usage guards who drive the offense
- Cluster 2 - Traditional Centers and Paint Anchors (38 players, about 11.8 million dollars) – rebounding and rim protecting bigs
- Cluster 0 - Versatile Forwards and Dynamic Bigs (97 players, about 7.8 million dollars) – rotation forwards and small ball bigs
- Cluster 1 - Complementary Wings and Shooting Guards (112 players, about 7.6 million dollars) – role players and shooters



These archetypes provide context for the salary analysis, since a fair contract for a role player looks very different from a fair contract for a franchise star.

Modeling

Three regression models were evaluated to predict log salary: Linear Regression as a baseline, and Random Forest and XGBoost as nonlinear tree based models. Model type was first compared using five fold cross validated R squared, where Linear Regression scored 0.59 and Random Forest scored 0.67. The stronger tree based result reflects that pay for production is nonlinear, with elite production paid at a steep premium and the CBA imposing hard floors and ceilings that a straight line cannot capture.

Random Forest and XGBoost were then tuned with GridSearchCV. The best Random Forest used 500 trees with a maximum depth of 10, and the best XGBoost used 200 trees with a learning rate of 0.1 and a maximum depth of 5. The table below reports each model on the held out test set. All metrics are on the log salary scale.

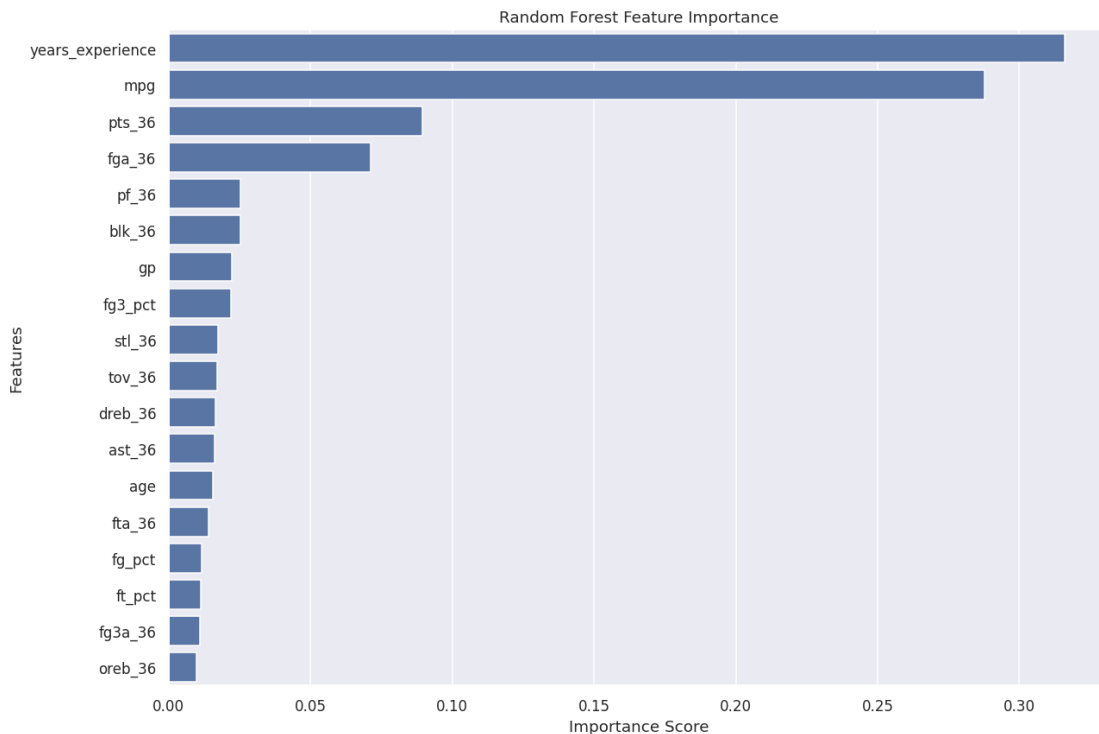
Model	Test MAE	Test MAPE	Test R ²
Linear Regression (baseline)	0.565	3.59%	0.618
Random Forest (final model)	0.581	3.73%	0.625
XGBoost	0.580	3.68%	0.624

Model performance on the held out test set. Linear Regression was scored on the same split to complete the comparison.

Random Forest was selected as the final model. Its R squared was the strongest, and it was carried forward into the feature importance and player value analysis. On the single held out test split the three models are close, which is expected given the small test set of 73 players, so the more stable cross validated result guided the choice. An R squared of about 0.62 means the final model explains roughly 62 percent of the variance in log salary.

Feature Importance

The chart below shows the built in feature importances from the final Random Forest model, which measure how much each feature reduced prediction error across the trees. The scores sum to one.



Random Forest feature importance. Years of experience and minutes per game dominate.

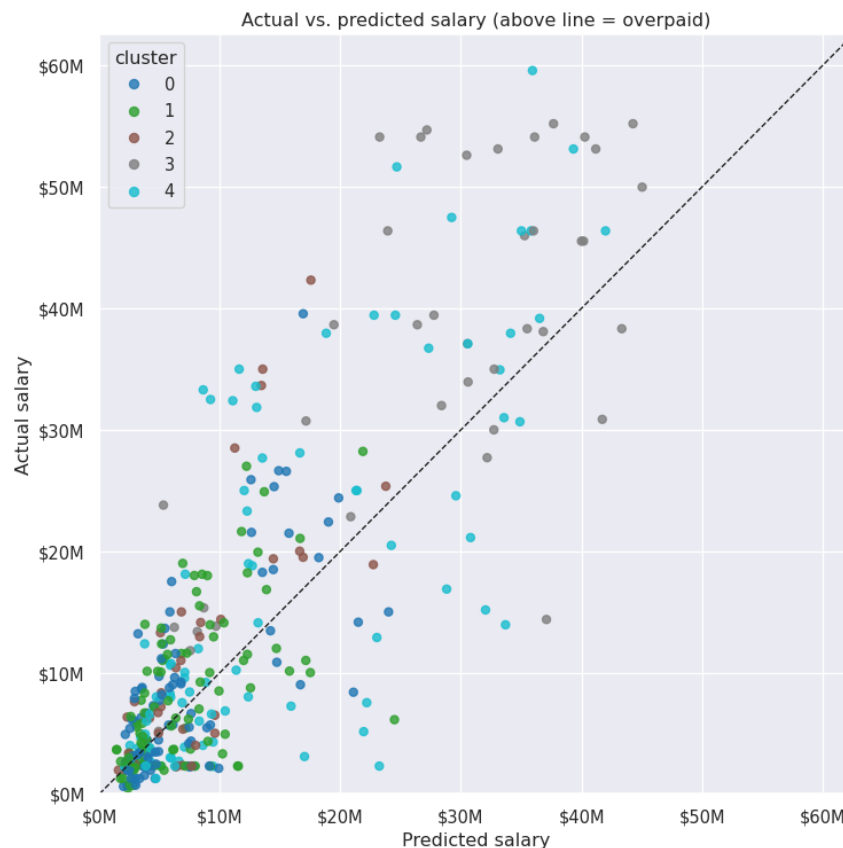
Two features dominate. Years of experience is the single most important feature, followed closely by minutes per game. Together they account for the large majority of the predictive power, while individual per 36 production stats such as points and field goal attempts matter far less.

This result has a clear interpretation. NBA pay is heavily structured by tenure, since rookie scale contracts are fixed for a player's first seasons and the maximum salary itself is defined in experience tiers. Minutes per game captures how central a player is to the team. Because the production stats were normalized to per 36 minutes, they no longer carry volume information, and that workload signal is instead captured by minutes per game. The production rate stats are also correlated with one another, so no single one stands

out as indispensable even though offense as a group still matters. In short, the market prices seniority and role more visibly than per minute production.

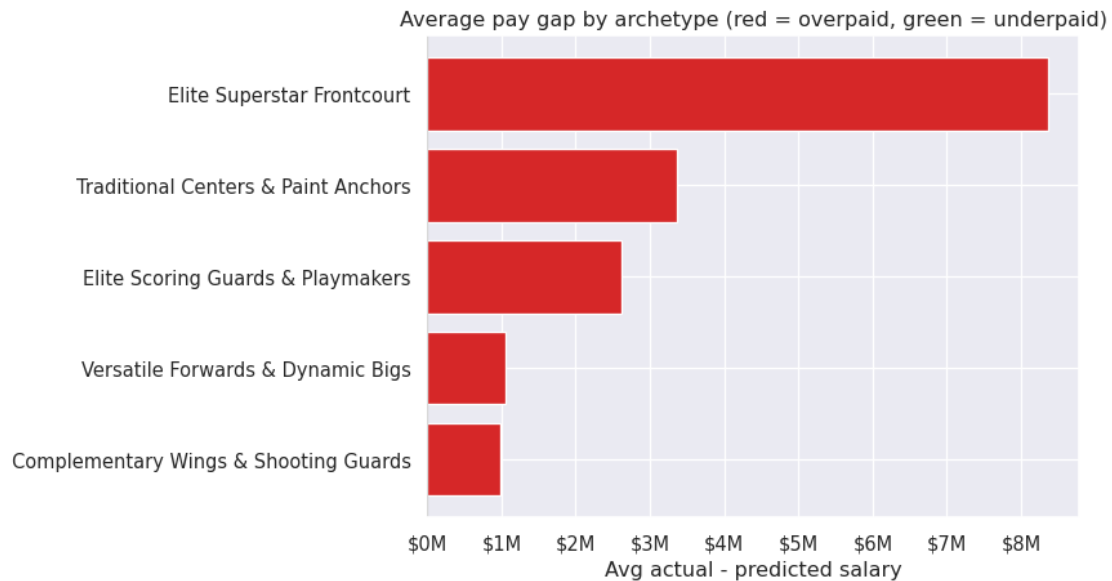
Player Value: Overpaid and Underpaid

To find overpaid and underpaid players, the final model predicted each player's salary using out of fold predictions, so every player is scored by a model that did not train on them. The gap between actual and predicted salary flags who is paid above or below their production based value.



Actual versus predicted salary. Points above the line are paid more than the model predicts.

The most overpaid players are almost all maximum contract stars, including Jimmy Butler, Kevin Durant, Giannis Antetokounmpo, and Paul George. Their production is elite, but the maximum salary sits above what any production based model will predict, so the highest earners naturally appear overpaid. The most underpaid players are productive names on inexpensive deals, including Austin Reaves, Deni Avdija, and Nickeil Alexander-Walker, many of them on rookie scale or minimum contracts.



Average pay gap by archetype. Overpayment rises with the tier of the group.

Grouping the gap by archetype shows that overpayment rises with the tier of the group. The Elite Superstar Frontcourt is the most overpaid archetype, followed by traditional centers and scoring guards, while the two role player clusters are the most efficiently paid. One caveat is that converting predictions from the log scale back to dollars adds a mild upward bias to every dollar gap, so the ordering of players and archetypes is the reliable signal rather than the exact dollar amounts.

Recommendations

The following recommendations follow directly from the feature importance and player value results.

1. Target productive players on inexpensive contracts. The value analysis flags players who produce well above what their pay predicts, such as Austin Reaves, Deni Avdija, and Nickeil Alexander-Walker. Rookie scale and bargain veteran deals are where the most surplus value sits, and pursuing these players in trades or extensions is the clearest path to roster value under the cap.
2. Apply discipline when paying for experience and minutes. Years of experience and minutes per game are the two dominant salary drivers, and the most overpaid players are established veterans whose pay reflects seniority more than current per minute production. Long maximum extensions for aging stars carry the largest risk of paying for past performance and should be weighed carefully against younger production.
3. Benchmark contracts within archetypes, not across the whole league. Comparing a player's pay to peers in the same archetype gives a fairer reference point than a league wide average. The Elite Superstar Frontcourt group carries the largest premium, while role player clusters are priced efficiently, which helps set realistic expectations in negotiations.

Ideas for Further Research

Several extensions could strengthen the analysis. Adding contract context such as years remaining, contract type, and injury history would separate current production from guaranteed money. Modeling several seasons together would improve stability and help untangle the fact that better paid stars also

tend to play more minutes. Applying a smearing correction to the log to dollar conversion would make the dollar level value gaps more accurate. Finally, adding advanced metrics beyond box score rates, such as efficiency and hustle plays, could capture value that traditional counting stats miss.

Conclusion

This project applied a full data science workflow to 2025-26 NBA data to predict player salary and identify market value. After wrangling and combining stats, salaries, and physical data for 582 players, a Random Forest model predicted log salary with a test R squared of about 0.62. The strongest drivers of pay were years of experience and minutes per game, well ahead of per minute production, which shows that the market prices tenure and role heavily. Using the model to compare predicted and actual pay surfaced the league's most overpaid and underpaid players and showed that superstar archetypes carry the largest premium while role players are paid efficiently. These results give a front office a data supported starting point for contract, trade, and roster decisions.